0x45.dev

BUILD LOG #2 · STACK

The $0 AI video stack

Every tool behind the 0x45.dev videos: an AI avatar with my cloned voice, captions, music and editing. **All free, all running on my laptop.**

$0 / MONTH

10 TOOLS

NO CREDITS



PAID VS LOCAL

No credits per second

TYPICAL AI VIDEO TOOL

$$$

credits for every second · monthly plan

THIS STACK

$0

runs on your laptop · re-render free

~40	~20	0
min for a 1-min video (M1 Max)	min of that is the AI face	credits used

💡 The catch is time, not money: start the render and walk away. Every voice change needs a new AI face, so fix the script first.



1 · 2

Voice and face

Chatterbox MIT

Clones your voice from a 10-15 s clip of a voice note. Write numbers and names the way you say them.

github.com/resemble-ai/chatterbox

DeepFilterNet MIT/Apache

Removes the hiss from the cloned voice, even while you speak.

github.com/Rikorose/DeepFilterNet

JoyVASA MIT

Turns the audio into head, eye and lip motion.

github.com/jdh-algo/JoyVASA

LivePortrait MIT

Renders your one image moving with that motion.

github.com/KlingAIResearch/LivePortrait



3 · 4

Captions and music

Whisper MIT

Word-by-word timings for the captions, and a check on every voice take.

github.com/openai/whisper

Rhubarb Lip Sync MIT

Mouth shapes from audio: the fast draft option for image avatars.

github.com/DanielSWolf/rhubarb-lip-sync

FluidSynth + GeneralUser GS free

The lo-fi music is written as code (MIDI) and played by this free synth and soundfont.

github.com/FluidSynth/fluidsynth



5 · EXTRAS

Edit it with code

Remotion free for individuals

Video as React code: layout, animation, captions, render to MP4. Free for individuals and small companies; bigger companies need a licence.

remotion.dev

FFmpeg LGPL/GPL

Cuts, encodes and masters the audio to -14 LUFS.

ffmpeg.org

Gemini app free tier

Makes the 3D cartoon look of you from your photos.

gemini.google.com

0x45.dev ✦ AI

Want the avatar steps too? Comment **AVATAR** on Build log #1.

That's the whole stack. Go build!

